

TSUDepth: Exploring temporal symmetry-based uncertainty for unsupervised monocular depth estimation[☆]

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ABSTRACT

When faced with occlusions and non-rigid motions, machines often struggle with depth estimation, a task effortlessly performed by humans with just one eye. Continuous RGB images embody rich temporal features, such as symmetry and optical flow, which current deep-learning models fail to effectively leverage. In response to this limitation, we introduce an innovative framework known as Temporal Symmetry-based Uncertainty (TSU)-Depth, aimed at enhancing the accuracy of unsupervised monocular depth estimation. The Temporal Symmetry-based Occlusion Optimization (TSOO) component plays a pivotal role in robustly identifying occluded regions and comparable optimization across adjacent frames. Simultaneously, we propose Temporal Optical Flow Masking (TOFM) to effectively identify and exclude static pixels (such as out-of-range depths and non-rigid objects) between adjacent frames. Additionally, we introduce Cross-Resolution Distillation (CRED) to enhance depth estimation accuracy across various resolutions, especially in low input resolution scenarios. Furthermore, we designed a new depth estimation structure utilizing the DPT structure and incorporating a GRU module to enhance performance details. Through extensive experiments on benchmark datasets, including KITTI, Cityscapes, and Make3D, our TSUDepth framework has consistently demonstrated state-of-the-art performance. Code is available at <https://github.com/BlueEg/TSUDepth/>.

1. Introduction

Depth is among the most important factors in navigation robots and autonomous vehicles. The AI can learn about a scene's geometry and plan a path while avoiding obstacles. Optical cameras, millimeter-wave radar, and LiDAR are crucial for obtaining depth information. However, as the cost of these tools decreases, so does their reliability. Some people and companies believe that optical cameras can provide reliable distance perception [1], just as humans judge distance with their eyes [2]. The ability to judge distance in visible light is complicated by occlusions and non-rigid motions. Predicting depth from binocular disparity is just one of the basic mechanisms of humans. Humans can also extract additional temporal and semantic information from visual images to aid depth perception, such as recognizing object categories and estimating object volumes, focusing on single targets, and understanding perspective relationships. Therefore, unsupervised monocular depth estimation is a challenging task for machines.

The pioneering work of Monodepth2 [3], as described in Section 3, is based on two fundamental assumptions to address the challenges

of occlusions and stationary pixels. These assumptions have laid the foundation for several subsequent works [4–7] in this field. The first assumption is that the occluded objects in the source frame are unlikely to find a correspondence in the target frame, leading to a high photometric error penalty. Therefore, occlusions can be found by comparing the loss of the corresponding area, which is called Per-Pixel Minimum Reprojection Loss [3]. The second assumption is that the pixel reprojection error in the warped image is lower than that of the original, so stationary pixels can be masked out by comparing the pixel reprojection error of the warped and original images, which is called Auto-Masking Stationary Pixels [3]. However, the strictness of these assumptions can result in the target area not being fully established, as shown in Fig. 3, so more robustness is needed. We believe that more semantics, such as temporal symmetry information between the preceding and subsequent frames, optical flow information between adjacent frames, and detailed information in high-resolution visible-light images, can be extracted to assist in estimating depth information.

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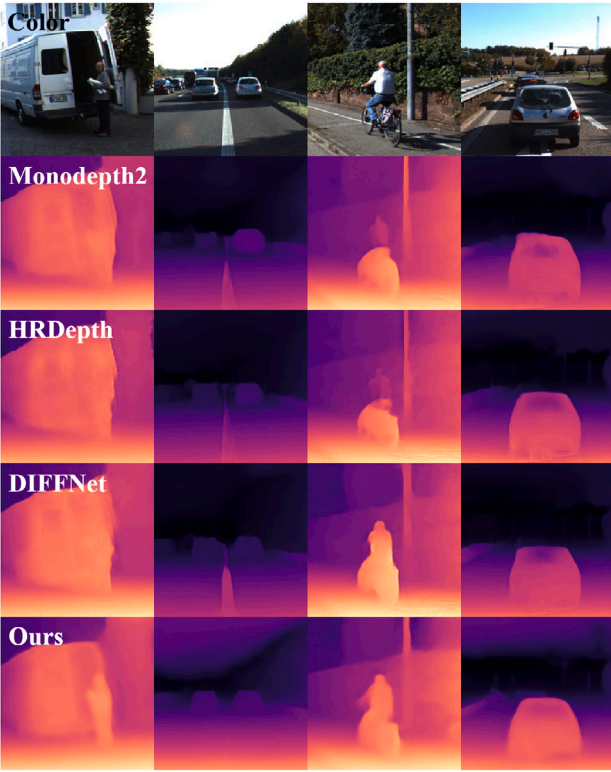


Fig. 1. Unsupervised monocular depth estimation from a single image. TSUDepth exhibits an exceptional ability to accurately predict detailed structures in depth maps (e.g., pedestrians and vehicles) regardless of illumination conditions (e.g. shades and shadows).

In our experiments, we observed that when occlusion happens because of objects in the scene, the occluded areas often lack sufficient information for accurate projection onto the target frame. Nonetheless, due to the opposite directions of projections from neighboring frames, these occlusion regions demonstrate symmetry concerning the target objects. This symmetry facilitates the exchange of complementary information between frames. Building on this observation, we have proposed an additional learnable algorithm based on temporal symmetry to enhance the robustness of unsupervised monocular depth estimation. First, we assert that the occluded areas in the source image are crucial contributors to the reprojection photometric loss. However, the information from these occluded regions can be complemented by using temporal symmetry across frames. Therefore, drawing inspiration from networks that learn data uncertainty, such as Ning et al. [8] and Zhu et al. [9], we introduced an uncertainty learning network. We further refined this proposal to allow the model to identify and improve certain regions in related frames while discarding uncertain regions. This design, characterized by its delicacy, has demonstrated outstanding results. Secondly, to address stationary pixels (no motion, making depth determination challenging) in related frames, we propose predicting temporal optical flow to identify stationary pixels between the source and target frames. This addresses a persistent misjudgment issue observed in Auto-Masking [3], as illustrated in Fig. 3.

The unsupervised monocular depth estimation of adjacent frames is based on the fundamental assumption that it is possible to calculate the scene depth from the movement of pixels in the scene. However, it is challenging to predict depth from images with minimal pixel movement accurately. Our requirements for depth are not high-resolution but must be sufficiently accurate. Traditional methods usually reduce the input frame's resolution by interpolation. However, the sampling operation of interpolation can cause the pixel movement to reduce or deform (width-height ratio change), affecting the depth estimation

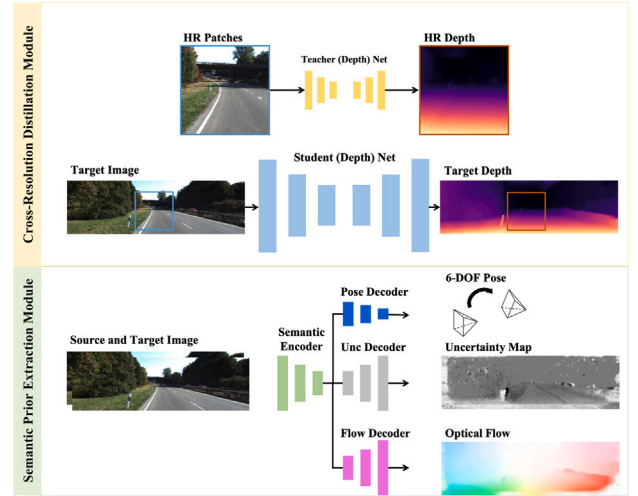


Fig. 2. Illustration of TSUDepth framework. The TSUDepth is composed mainly of two modules. The first is the Cross-Resolution Distillation Module, which utilizes a Student Depth Net to predict the output depth and a Teacher Depth Net to extract a high-resolution (HR) depth patch used to distill parts of the output depth as indicated by the red rectangle. The second module is the Semantic Prior Extraction Module, which is composed of a semantic encoder and multiple decoders (Pose, Optical Flow and Uncertainty Decoder).

accuracy. Therefore, we propose using full-resolution patches in the original to distill the low-resolution depth simultaneously as a kind of random attention supervision. This distillation-based method improves the prediction's overall accuracy without increasing the network's input resolution.

Researchers who have followed Monodepth2 [3] have improved the performance of monocular depth estimation by improving the structure of the network [4,5,7,10], achieving the current state-of-the-art (SOTA). They have found that sharing more scale characteristics, such as HRNet [11] and attention structures [5,7], can improve performance in general. Inspired by Dense Prediction Transformer (DPT) [12,13], we introduced DPT as an attention-based encoder [14] in unsupervised monocular depth estimation, allowing the network to capture long-range dependencies between image features, and designed a GRU-based depth decoder that can make full use of the features of the DPT encoder, achieving the new SOTA result, as shown in Fig. 1. In summary, the main contributions of our work are as follows:

- To tackle the challenges posed by occlusion and inadequate Minimum Reprojection Loss, as well as to leverage symmetric information from adjacent frames, we propose a multiframe optimization method based on uncertainty learning estimation. This method aims to extract valid information from related frames' photometric errors while disregarding invalid information. We refer to this approach as the Temporal Symmetry-based Occlusion Optimization (TSOO) method.

- To tackle stationary pixels and out-of-range points, we replaced the unstable Auto-Masking in Monodepth2 [3] with Temporal Optical Flow Masking (TOFM), which employs unsupervised optical flow, enhancing stability and effectiveness in removing stationary pixels and preventing overfitting. Additionally, we introduced the Cross-Resolution Distillation Method (CRED) to enhance depth estimation accuracy across resolutions, especially addressing accuracy degradation with decreasing input resolution.

- We introduced the Dense Prediction Transformer (DPT) model for unsupervised depth estimation tasks. Deliberately designing the decoder, we integrated a focused GRU module to enhance long-term spatio-temporal correlations, resulting in the DPT-G model. We named our collection of methods TSUDepth, and through experiments, demonstrated their advanced capabilities across various model structures, input resolutions, and datasets.



Fig. 3. Comparison of our proposed methods with Monodepth2 [3]. Auto-masking and TOFM identify stationary pixels across frames, but Auto-masking may mislead in texture-less regions, while our TOFM method achieves greater precision through optical flow integration. MiniRL can identify occluded areas (in black) within the mask. However, MiniRL is discrete and prone to significant misjudgments, while our approach is continuous, precise, and provides a more accurate representation.

2. Related works

2.1. Unsupervised depth estimation

Unsupervised depth estimation is a rapidly evolving field of research in machine learning that aims to estimate the distance from stereo views or video sequences. Godard et al. [15] proposed the loss of left-right disparity consistency to estimate depth from stereo visual matching. Furthermore, Zhou et al. [16] are the pioneers who proposed simultaneously estimating depth and pose with different networks. Inspired by them, Godard et al. [3] designed a minimum reprojection loss and an auto-masking loss, establishing the basic framework of unsupervised monocular depth estimation. Many tasks [4–7,17] improve accuracy by reforming the network structure. PackNet [17] uses symmetrical packing and unpacking blocks to propagate dense appearance and geometric information maximally. CAdDepth [5] introduces channel-wise attention to the depth estimation network, which emphasizes channel feature representation. HR-Depth [4] uses HRNet [11] that can make full use of the resolution information of the feature to replace the original U-Net decoder. DIFFNet [7] introduces HRNet to the depth encoder, which uses semantic information in the down and upsampling procedures. Moreover, RA-Depth [6] uses HRNet as an encoder and decoder at the same time. Jung et al. [18] and Chen et al. [19] have designed loss functions relying on semantic segmentation information. In contrast, our approach delves into additional semantic information embedded in RGB images, including optical flow and uncertainty.

2.2. Uncertainty learning in dense prediction

Uncertainty estimation was originally designed with Bayesian neural networks (BNN) in the early articles [20,21]. Later, Kendall et al. [22] proposed the aleatoric uncertainty of the noise in the observations. The epistemic uncertainty that explains the uncertainty of the model was studied in a joint framework in a pixel-wise depth regression task. And then more researchers [23–25] promoted uncertainty estimation to filter out noisy predictions within the network. Yang et al. [26] modeled the photometric uncertainties of pixels in the input images in unsupervised monocular depth estimation. Wong et al. [27,28] developed adaptive regularization and data fidelity with the same idea of uncertainty to improve hard boundary prediction in

unsupervised depth completion. Furthermore, in our work [9], uncertainty was also an inference from Bayesian methods, and they proposed a novel two-stage method to reduce uncertainty in a supervised depth completion task.

2.3. Optical flow in depth estimation

Optical flow is an essential component in video processing, as it enables the detection of warping information between related frames by predicting the movement of pixels [29,30]. Supervised optical flow estimation is based on a synthetic optical flow dataset with flow labels [31]. Unsupervised methods, on the other hand, are similar to standard unsupervised depth estimation based on minimum reprojection photometric error [30]. Zou et al. [32] proposed an unsupervised framework to predict depth and optical flow, which uses depth and pose to calculate 2D optical flow supervised by the predicted flow. More recently, Zhu et al. [33] used a pre-trained Flow-Net to remove out-of-range depths in stereo-based unsupervised depth estimation.

2.4. Transformer in depth estimation

The Transformer [14] is a robust neural network architecture that efficiently models long-range dependencies, making it a versatile and practical tool for a wide range of domains and tasks. Additionally, Dosovitskiy et al. [34] first introduced the Transformer model for image classification. The main advantage of the Vision Transformer (ViT) model lies in its ability to model long-range dependencies between pixels or regions in an image without relying on convolutional operations. Ranftl et al. [12] reconstructed the Transformer decoder using convolutional networks and maintained the image resolution of the tokens. Lee et al. [35] improved the transformer's capabilities by collecting tokens of different sizes to achieve different scales and receptive fields. Zhang et al. [36] introduced a lightweight architecture that combines CNNs and Transformers efficiently, achieving results comparable to existing approaches.

2.5. Comparison with relevant methods

Uncertainty-Driven Loss [9]: Our TSOO method, inspired by uncertainty-driven loss [9] in dense prediction, distinguishes itself by utilizing temporal symmetry from both preceding and subsequent frames with respect to the target frame. In contrast to approaches that modify weights in the loss map, our dynamic reweighting of reprojection errors, guided by uncertainty map comparisons across frames, effectively addresses challenges such as occlusion during training.

UnOS [37]: TOFM and UnOS play distinct roles in their respective tasks. TOFM enhances depth estimation for stationary objects by focusing on unsupervised pixel-wise optical flow, excluding minimal motion pixels. In contrast, UnOS optimizes global pose by comparing unsupervised optical flow with rigid flow from depth and pose. However, UnOS faces challenges in accurately determining the depth of stationary objects, while TOFM effectively addresses this by excluding stationary pixels using minimal values in unsupervised flow. Moreover, TOFM achieves efficiency through simultaneous training, while UnOS adopts pre-training of optical flow and joint training of depth with two independent networks for pose and optical flow.

RA-Depth [12]: RA-Depth focuses on adapting a unified model for diverse image resolutions, improving depth estimation across various resolutions using cross-scale Depth Consistency Loss. In contrast, our Cross-Resolution Distillation (CRED) method trains low-resolution depth models using a high-resolution (HR) teacher network. RA-Depth resizes images to fixed resolutions (L, M, H) for depth prediction, while CRED enhances performance by randomly sampling high-resolution patches, minimizing computational overhead while leveraging high-definition information at the original resolution.

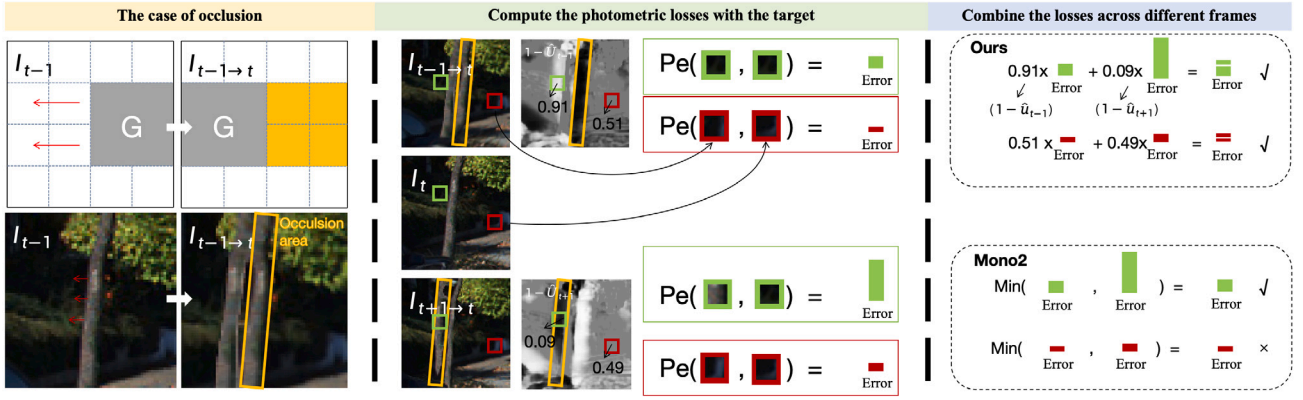


Fig. 4. We propose a Temporal Symmetry-based Occlusion Optimization (TSOO) for symmetric occlusion handling. On the left, as the occluding object G moves from right to left, it leaves an uncovered orange area in $I_{t-1 \rightarrow t}$, which cannot be predicted from I_{t-1} . In the middle, we calculate the error in the red and green boxes and observe significant errors in the occluded area. On the right, we compare the Per-Pixel Minimum Reprojection Loss and our TSOO method. In non-occlusion areas, the Monodepth2 approach may choose the minimum error in the source, resulting in a loss of some supervised information. By contrast, our method retains the source reprojection errors and avoids the impact of occlusions, leading to better performance.

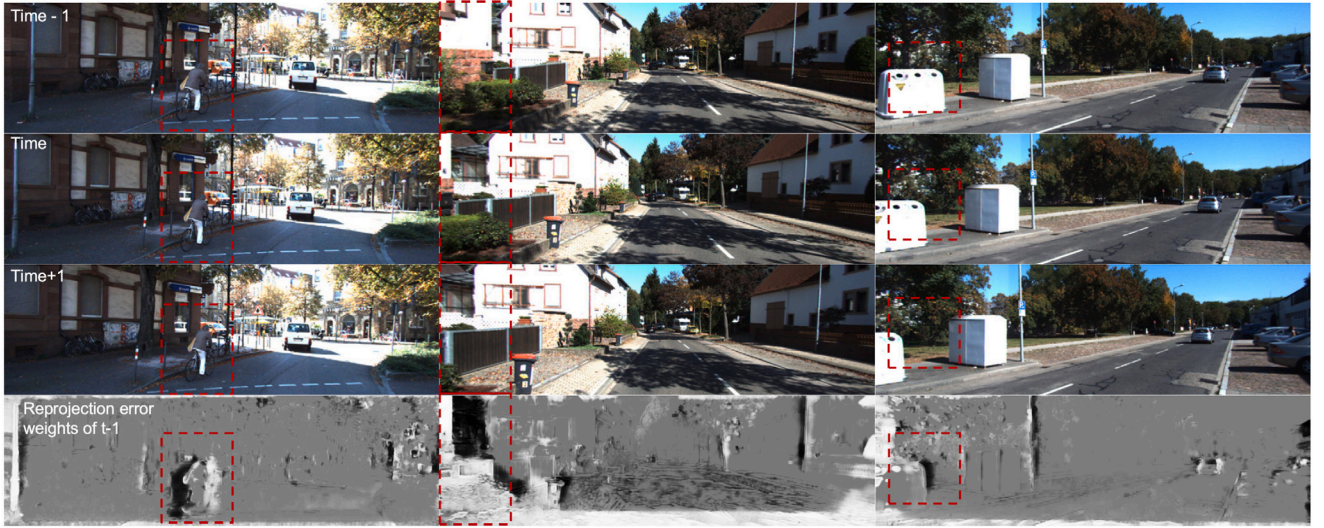


Fig. 5. Examples of Uncertainty-Driven Weights: Focusing on the middle column—while our vehicle advances, the left wall retreats, creating a significant uncertain zone (depicted in black in the third row) previously obscured by the wall. Conversely, when no white wall appears at time $t+1$, its transformation towards time t cannot be restored, resulting in elevated weights (depicted in white). A similar analysis applies to pedestrians and vehicles in the other columns. Our proposed method aims to detect such occlusions and mitigate their impact on the loss.

3. Methodology

We will begin by reviewing traditional settings for unsupervised monocular depth estimation. Then, we will introduce TSUDepth, a framework that integrates three supporting modules: TSOO (Temporal Symmetry-based Occlusion Optimization; see Fig. 4), TOFM (Temporal Optical Flow Masking; see Fig. 9), and CRED (Cross Resolved Distillation; see Fig. 8). Additionally, we introduce the DPT-G model, a robust transformer architecture equipped with an efficient GRU decoder (depicted in Fig. 11). The entire architecture of our framework is illustrated in Fig. 2.

3.1. Preliminaries

Unsupervised depth estimation involves calculating the depth and relative pose to create a novel view. Firstly, use a Depth net to input the target image I_t and obtain the corresponding depth map D_t , and a Pose Net to input the source image I_s and target image I_t and obtain

the relative pose $P_{t \rightarrow s}$. With these outputs, we can project the source I_s to the target I_t using the depth D_t , the pose $P_{t \rightarrow s}$, and the intrinsic matrix of the camera K .

$$I_{s \rightarrow t} = I_s \langle \text{proj}(D_t, P_{t \rightarrow s}, K) \rangle \quad (1)$$

During training, Monodepth2 [3] minimizes the following photometric reprojection error.

$$pe(I_t, I_{s \rightarrow t}) = \alpha(1 - SSIM(I_t, I_{s \rightarrow t})) + (1 - \alpha)\|I_t, I_{s \rightarrow t}\|_1 \quad (2)$$

and chooses $s \in \{t-1, t+1\}$ simultaneously, so **Per-Pixel Minimum Reprojection Loss(MiniRL)**[3] can be written as:

$$\mathcal{L}^{Depth} = \min_s pe(I_t, I_{s \rightarrow t}) \quad (3)$$

and the **Auto-Masking Stationary Pixels**[3] is defined by:

$$\mu = [\min_s pe(I_t, I_{s \rightarrow t}) < \min_s pe(I_t, I_s)] \quad (4)$$

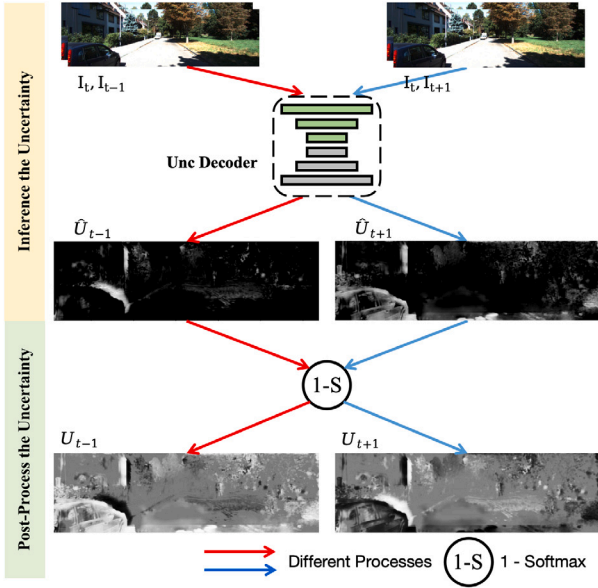


Fig. 6. Predicting uncertainty $\hat{U}_s$, $s \in \{-1, 1\}$ across frames $\{I_t, I_s\}$ and generating weights U_s for loss combination with softmax. By comparing the weights U_s with the uncertainties $\hat{U}_s$, we can observe that in regions with high uncertainty in a frame, the proportion of the loss weight there will be low.

3.2. Temporal symmetry-based occlusion optimization

Occlusion is the main challenge in aligning video sequence images, which can produce uncertain regions after warping (see Fig. 4) regardless of whether it is based on optical flow or pose with depth for reprojection. Per-Pixel Minimum Reprojection Loss in Monodepth2 [3] addresses this problem by selecting the smaller loss value, corresponding to the more reliable pixel in the final loss, optimizing the frame with minimal loss among multiple frames. However, it is important to note that this approach is only appropriate for occlusion areas and may result in abandoning more supervision information in more complex regions. At the same time, it also does not conform to the principle of multi-view fusion in structure-from-motion (SfM) [38]. As a result, edges and details will appear unclear in depth. We believe that we can learn such areas from the distribution of occlusion areas, although mature algorithms for detecting occlusion areas with significant and inevitable loss values are lacking. Inspired by data uncertainty learning [8,9,22], which actively learns the distribution of difficult areas, we propose using a network that produces an uncertainty map to reweight the final loss. In our previous work [9], we have used the following joint optimization loss function:

$$\frac{1}{N} \sum_{i=0}^{N-1} \left(e^{-\hat{u}_i} (\hat{x}_i - x_i)^2 + 2\hat{u}_i \right) \quad (5)$$

where $\hat{u}_i$ ($\hat{u}_i > 0$) is the uncertainty of the pixel i and $\hat{x}_i, x_i$ is the prediction and ground truth of the pixel. N represents the number of pixels. The loss function is typically defined as a weighted sum of errors between the predicted values $\hat{x}_i$ and the ground truth values x_i . To account for the uncertainty of each prediction, we introduce a per-pixel uncertainty term $\hat{u}_i$ such that the difficult region will be assigned to a high uncertainty value, resulting in a small loss weighed by $e^{-\hat{u}_i}$. At the same time, the second term of the formula $2\hat{u}_i$ is regularization from the sparse priority of uncertainty, which is designed to limit the growth of uncertainty values and prevent overfitting [9].

In this work, we adapt this loss function to unsupervised depth estimation, where we calculate the re-projection photometric error between two frames $I_{s \rightarrow t}$ and I_t . To adapt the loss function for this

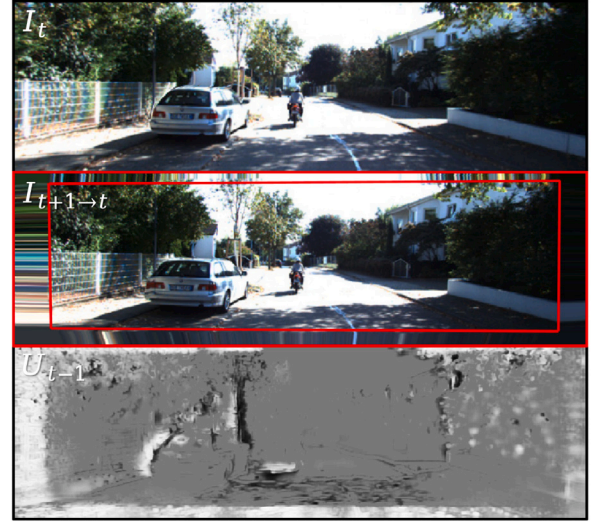


Fig. 7. An illustration of occlusion caused by view shrinking: Pixels left uncovered along the image border result from the camera zooming out. This phenomenon consistently appears in $I_{t+1 \rightarrow t}$, resembling the middle image. Our proposed TSOO method can automatically detect such regions, enabling $pe(I_t, I_{t-1 \rightarrow t})$ to be weighted more heavily, reflected as high-lighted areas along the edges of U_{t-1} .

task, we rewrite Eq. (5) as follows:

$$\frac{1}{N} \sum_s \sum_{i=0}^{N-1} \left(e^{-\hat{u}_{i,s}} pe_i(I_t, I_{s \rightarrow t}) + 2\hat{u}_{i,s} \right), s \in \{t-1, t+1\} \quad (6)$$

where pe denotes the photometric reprojection error mentioned previously, $s \in \{t-1, t+1\}$ represents the source images. In an unsupervised setting, there are two main differences from supervised dense prediction. First, no ground-truth depth is available for us to use as supervision. The network's predicted disparity (depth) is only a tool for warping with the pose, which leads to uncertainty in the formula that does not belong to depth but a re-projection error map. Second, we use the results of multi-frame visual warping ($I_{s \rightarrow t}, s \in \{t-1, t+1\}$) to supervise by the target frame. This allows us to generate additional information about occlusion areas from different transformation directions (i.e., backward and forward), thereby highlighting temporal symmetry. And this is similar to how humans can avoid occlusions in vision by shaking their heads. In previous work such as sfmlearner [3, 16], the average or minimum error of different frames has been used, where errors are combined with different weights ([0.5 0.5][16], [0 1][3]) that sum up to unity. We propose combining these errors with an uncertainty-based weights U to achieve better complementary information. Specifically, we replace the first item in Eq. (6) with a simple uncertainty form for $\{1 - x | x \in [0, 1]\}$ that has an approximate value domain and a gradient direction of $\{e^{-x} | x > 0\}$.

To combine photometric loss across different frames, we specify $\sum_s (1 - \hat{u}_{i,s}) = 1, s \in \{t-1, t+1\}$. Hence, we combine the output uncertainty map $\hat{U}_s$ ($\hat{U}_s \geq 0$) with softmax: $\{U_{t-1}, U_{t+1}\} = 1 - \text{Softmax}\{\hat{U}_{t-1}, \hat{U}_{t+1}\}$, to ensure that $\sum_s u_{i,s} = 1$. The final joint optimization function is written as:

$$\mathcal{L}^{UD} = \frac{1}{N} \sum_s \sum_{i=0}^{N-1} u_{i,s} pe_i(I_t, I_{s \rightarrow t}) + \alpha \hat{u}_{i,s}, \sum_s u_{i,s} = 1 \quad (7)$$

With this formula, α was set to $1e^{-2}$, the re-projection errors of different source $pe(I_t, I_{s \rightarrow t})$ are re-weighted during the optimization process based on their corresponding uncertainty values U_s . As shown in Fig. 4, pixels with high uncertainty in source frame I_s receive a smaller weight for its re-projection error, while those with lower uncertainty in another frame receive a higher weight. To implement this, we designed a Data Uncertainty Decoder after the Semantic Encoder.

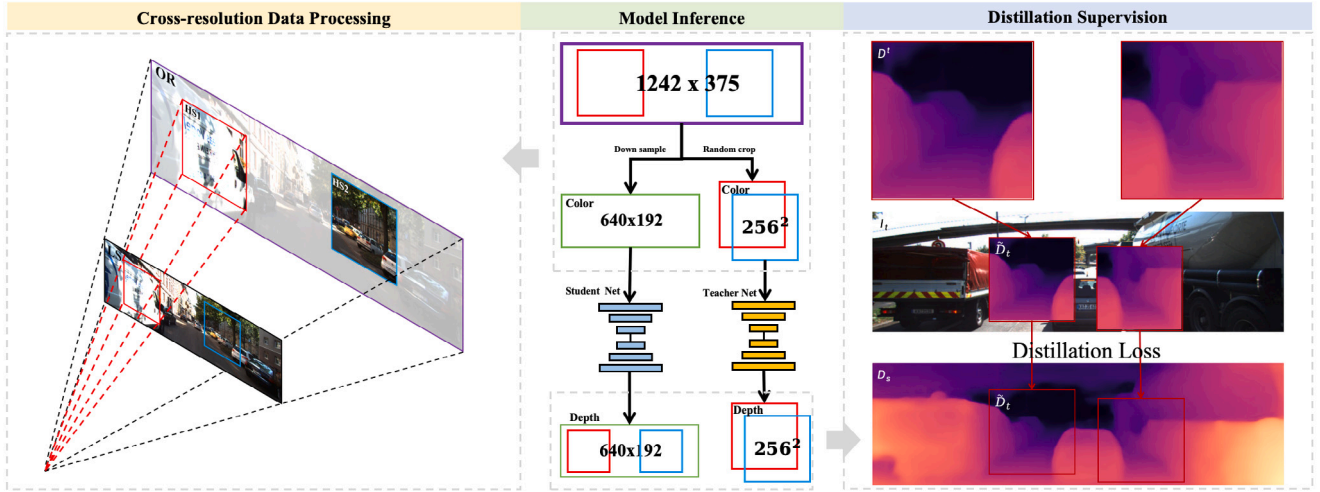


Fig. 8. The Cross-Resolution distillation method involves extracting high-resolution (HR) image patches from the original-resolution (OR) image, as shown on the left side of the diagram. These HR patches are then used to provide supervision for a lower-resolution input through TeacherNet, as shown on the right side of the diagram.

This decoder extracts features from the source frames I_s and the target frame I_t , and outputs the uncertainty values $\{\hat{U}_{t-1}, \hat{U}_{t+1}\}$. Then, obtain multi-frame reprojection error weights by comparing $\{U_{t-1}, U_{t+1}\} = 1 - \text{Softmax}\{\hat{U}_{t-1}, \hat{U}_{t+1}\}$, where $\hat{U}_s \geq 0$ is the output of the network, as shown in Fig. 6.

Our method is particularly effective for a specific type of occlusion, which we call a “view-shrinking occlusion”, as shown in Fig. 7. Assuming the vehicle moves forward, edge pixels will shrink inward and produce occlusion when the image is warped from the frame I_{t+1} to I_t . Our method can predict such regions more accurately than other competing methods, as shown in Fig. 4. and we present visualizations of reprojection error weights $\hat{U}_{t-1}$ from some TSOO outputs, shown in Fig. 5. As foreground objects move, extensive regions of uncertainty emerge in the backward direction, while corresponding regions of certainty become evident in the forward direction. If left unaddressed, the uncertainty regions resulting from occlusion can lead to blurry object boundaries in the estimated depth. Furthermore, our proposed TSOO method goes beyond merely identifying occluded regions and selecting the correct loss values among these frames based on uncertainty during training. It also incorporates a strategy of averaging the loss values from multiple frames in non-occluded regions, as illustrated by the red regions in Fig. 4. The uncertainty remains smooth in non-occluded areas, where losses from multiple frames do not escalate. Post-softmax, the confidence of pixels in these regions regresses to $[0.5, 0.5]$, mirroring the average in multiple frames and preserving information from each frame. In contrast, Monodepth2 adopts a simplistic approach by selecting the minimum value in occluded and non-occluded areas. Unfortunately, this results in the loss of nearly half of the valuable information across multiple frames, particularly evident in the red regions.

3.3. Temporal optical flow masking

The intuition is that pixels that remain stationary between frames are likely to be small in the estimated optical flow, as shown in Fig. 9. Using the relationship $I_s(x-u, y-v) = I_t(x, y)$, we can derive the optical flow $F_{s \rightarrow t}$, which facilitates the identification of stationary points. To identify stationary pixels in the optical flow of $I_s \rightarrow I_t$, we propose to threshold the distance of pixel displacement as follows:

$$M_{s \rightarrow t}^{TOFM} = [|F_{s \rightarrow t}| > \text{threshold}] \quad (8)$$

where $F_{s \rightarrow t}$ is the optical flow map from frame I_s to frame I_t . In addition to selecting static regions, the threshold parameter in the above equation serves two other purposes. First, it excludes static

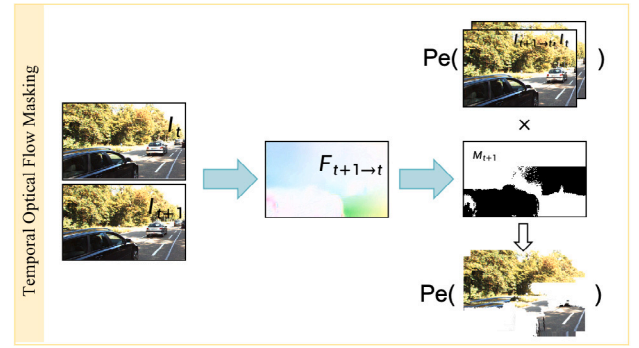


Fig. 9. We propose a Temporal Optical Flow Masking (TOFM) method to address the misclassification of non-rigid objects, such as vehicles moving at the same speed as the host vehicle or stationary pixels like the sky on the left, resulting in erroneous depth predictions. While the Monodepth2 approach employs the Auto-Masking method to eliminate stationary pixels, it may still fail to distinguish non-rigid objects from stationary ones, as shown in the middle. In the TOFM method, we use optical flow to identify and eliminate out-of-boundary and stationary pixels from the scene, as shown on the right.

pixels in related frames caused by non-rigid movement. To see how to ensure that the network can predict the depth of the masked areas correctly, we note that parked vehicles on the roadside often share similar features with masked vehicles. In this way, they provide valuable prior information required for masked areas. Second, unsupervised methods assume that the scene only has rigid motion. However, there are situations where the primary vehicle (with a camera) stops and waits, resulting in adjacent frames having little movement. Additionally, the intensity of optical flow is directly proportional to vehicle speed. Therefore, the thresholding strategy can exclude abnormalities arising from parking or speeding.

Specifically, we calculate the masked loss as $M^{TOFM} \cdot \mathcal{L}_{depth}$. Furthermore, we add a decoder to the semantic encoder to predict $F_{s \rightarrow t}$, from which we can get $I_{s \rightarrow t}^{flow} = \text{warp}(I_s, F_{s \rightarrow t})$ with the optical flow loss function simultaneously:

$$\mathcal{L}^{flow} = \frac{1}{N} \sum_{i=0}^{N-1} u_{i,s} p e_i(I_t, I_{s \rightarrow t}^{flow}), \quad (9)$$

where we have used the same TSOO errors weights in Section 3.2 to combine photometric errors from optical flow, which also addresses occlusion questions. With the Temporal Optical Flow Masking, the

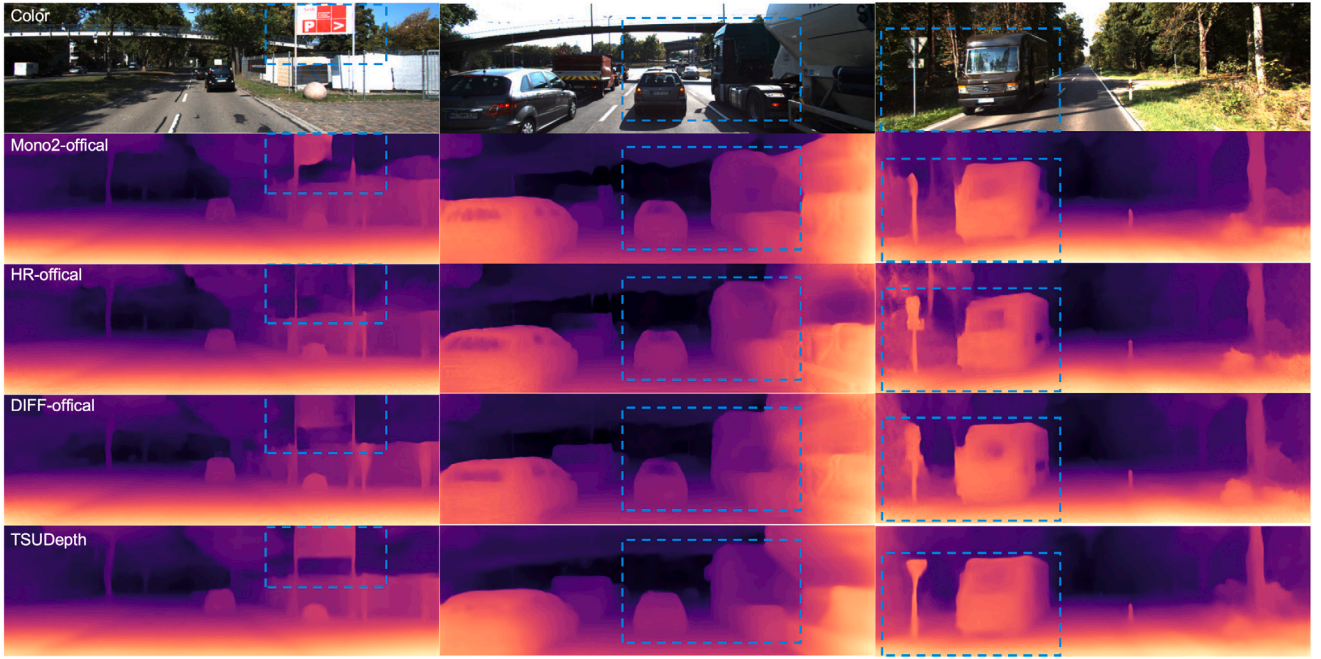


Fig. 10. Visualization of depth estimation results: The first row shows the input target frames, while the following rows display the depth estimation results using Monodepth2 [3], HR-Depth [4], and DIFFNet [7], respectively. The last two rows present the depth estimation results obtained using our TSUDepth framework, utilizing the depth prediction network structures of DIFFNet and our own designed DPT-G depth prediction network structure.

TSOO can be rewrite to:

$$\mathcal{L}^{UD} = \frac{1}{N} \sum_s \sum_{i=0}^{N-1} M_{s \rightarrow t}^{TOFM} \cdot u_{i,s} pe_i(I_t, I_{s \rightarrow t}) + \alpha \hat{u}_{i,s}, \sum_s u_{i,s} = 1 \quad (10)$$

3.4. Cross-resolution distillation method

To meet the real-time requirement of depth estimation, downsampling the original image is a popular strategy. For example, in KITTI data pre-processing, the original image of size 1242×375 is often down-sampled to 640×192 . However, the smaller the sampling size, the less accurate the estimated depth is, as shown in Fig. 12. To address this issue, we propose leveraging the dataset's maximum resolution to refine the sampled depth. Inspired by the cross-scale depth consistency loss in RA-depth [6], we adopted different networks to predict depths of different resolutions and supervise the consistency between them by patches. But unlike RA-Depth, we employ a distillation method in training. The teacher network takes the original resolution cropped patches as input. It generates corresponding depth patches that supervise the corresponding position depth patches in the student depth map, as shown in Fig. 8.

First, we interpolate the original resolution images I^{Ori} to the input size I^{Stu} of the student network and randomly crop $N(N=2$ in our experiments) image patches I^{Tea_i} from I^{Ori} . Second, we obtain the corresponding depth maps D_t^{Stu} and $D_t^{Tea_i}$ that follow the same $P_{s \rightarrow t}$ through the Student and Teacher Depth Networks. Finally, we interpolate $D_t^{Tea_i}$ to the same size $\tilde{D}_t^{Tea_i}$ as the corresponding patch $\tilde{D}_t^{Stu}$ in D_t^{Stu} . The cross-resolution distillation loss is calculated as follows:

$$\mathcal{L}^{CR} = \sum_{i=1}^2 pe(\tilde{D}_t^{Tea_i}, \tilde{D}_t^{Stu_i}) \quad (11)$$

In our implementation, two distinct high-resolution patches are applied for distillation, both utilizing a shared teacher depth network. The parameter α is set to 0.85.

We employed randomly sampling teacher patches to supervise the student, ensuring a limited computational load during training. This approach maintains a small input resolution during inference, resulting

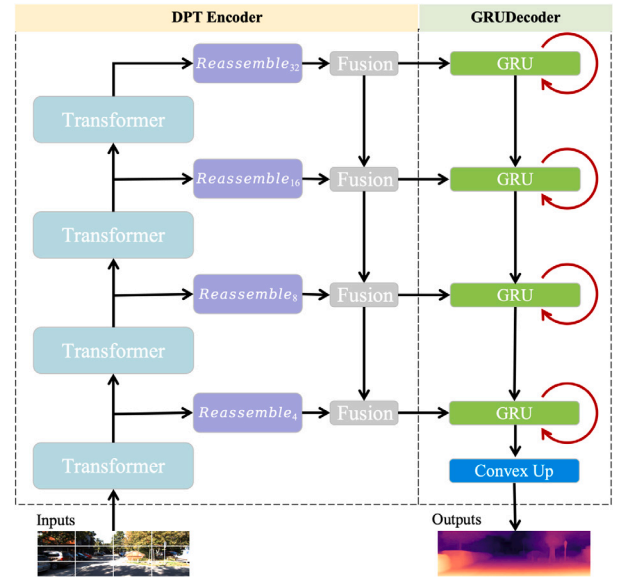


Fig. 11. Our DPT-G Depth Net. Our Depth Net utilizes an ImageNet pretrained DPT [12] Encoder. In addition, we incorporate GRU modules [10,39] updating the features iteratively to decode the depth and apply a convex up [39] module to output the depth map.

in fast inference speeds and high-depth estimation accuracy, as shown in Table 2 and Fig. 12.

3.5. DPT-G architecture

Our framework offers the flexibility to accommodate various existing depth estimation models [3,4,6,7], and in our experiments, we selected representative models including Monodepth2 [3], HRDepth [4], DIFFNet [7], and RA-Depth [6], as shown in Table 2. At the same time, inspired by the impact of the Transformer structure on visual

tasks, as demonstrated by Dosovitskiy et al. in the work on vision tasks [34]. Specifically, our aim is to employ the Transformer structure as the backbone for unsupervised monocular depth estimation. We harness its feature extraction and storage capabilities by leveraging the powerful multi-head attention. Given the ease of acquiring datasets for unsupervised tasks and their independence from specific experimental environments, we strive to extract more accurate real-world depth information.

We adopt the DPT-Hybrid in Dense Prediction Transformer (DPT) [12] as our backbone, which successfully integrates the Vision Transformer (ViT) structure into dense prediction tasks. However, a challenge arises because of the distinct architectures of supervised and unsupervised depth estimation models. While supervised models output depth directly, unsupervised models employ an encoder-decoder composite structure and output disparity [3]. Consequently, we extract the ViT structure of DPT as the Encoder and design a novel Decoder tailored to complement the Transformer Encoder.

To fully exploit the feature information in RGB images/features, recurrent networks are often an effective approach, enhancing specific information representation through repetitive structures. And such structural designs have already been applied in Yan et al. [40]. To enhance the original Monodepth2 Decoder, we introduced a Conv Gated Recurrent Unit (GRU) and Convex Upsampling [39], which facilitates smoother convergence in depth estimation while also conserving parameter space, as illustrated in Fig. 11. And the ConvGRU is defined as:

$$\begin{aligned} hx &= \text{torch.cat}([h, x], \text{dim} = 1) \\ z &= \sigma(\text{Conv2d}(hx)) \\ r &= \sigma(\text{Conv2d}(hx)) \\ q &= \tanh(\text{Conv2d}(\text{torch.cat}([r \odot h, x], \text{dim} = 1))) \\ h &= (1 - z) \odot h + z \odot q \end{aligned}$$

where σ represents the sigmoid function, $\odot$ represents element-wise multiplication, and **Conv2d** represents the 2D convolution operation with a kernel size of 3x3. Typically, we allow the feature h to take four recurrent iterations. Our model begins by extracting multiscale, strongly correlated global features using the DPT-Encoder. Subsequently, these features undergo further refinement and enhancement through a lightweight GRU-Decoder. Given the intricate training conditions inherent in Transformer networks, the DPT-Hybrid was trained by initiating the process with a Vision Transformer (ViT) model pre-trained on ImageNet.

3.6. Total training loss

Other Structures. We use the pose encoder from [3] as the Semantic Encoder. Our network architecture is divided into single-frame (Cross-Resolution Distillation) and dual-frame (Semantic Prior Extraction) modules. The Cross-Resolution Distillation Module is responsible for predicting Teacher and Student depth and distilling the parts of Student by Teacher Depth patches. At the same time, the Semantic Prior Extraction Modules extract inter-frame features by the Semantic Encoder and use different decoders to output inter-frame semantic predictions such as global pose, inter-frame uncertainty, and pixel-level optical flow. Furthermore, the uncertainty and optical flow decoder has almost the same structure as the depth decoder in Godard et al. [3], connected after the Semantic Encoder, as shown in Fig. 2.

Training Strategy. The Cross-resolution distillation(CRED) includes two relatively independent depth estimation networks (Teacher network and Student network). To avoid mutual influence, we have set up two independent optimizers. The teacher optimizer $\mathcal{O}_T$ and the student optimizer $\mathcal{O}_S$. The $\mathcal{O}_T$ is responsible for optimizing the generation of high-resolution depth blocks and optical flow, while the $\mathcal{O}_S$ focuses on generating full-size depth images and ensuring the consistency of cross-resolution depth estimation, as shown in Alg. 1. where ζ is set to 0.5

Algorithm 1 TSUDpth Training Optimization in one iteration

- 1: **Initialize:** $\mathcal{O}_T.\text{zero_grad}()$, $\mathcal{O}_S.\text{zero_grad}()$
- 2: **Calculate:** $\mathcal{L}_{Tea} = \mathcal{L}^{UD}(I_{s \rightarrow t}^{Tea_i}, I_t^{Tea_i})$
 $+ \mathcal{L}^{flow}(I_{s \rightarrow t}^{flow}, I_t)$
- 3: $\mathcal{L}_{Tea}.\text{backward}()$
- 4: **Optimize** $\mathcal{O}_T$: $\mathcal{O}_T.\text{step}()$
- 5: **Calculate:** $\mathcal{L}_{Stu} = \mathcal{L}^{UD}(I_{s \rightarrow t}^{Stu}, I_t^{Stu})$
 $+ \zeta \mathcal{L}^{CR}(\tilde{D}_t^{Stu}, \tilde{D}_t^{Tea_i})$
 $+ \beta \mathcal{L}_{sm}(D_t^{Stu}, I_t^{Stu})$
- 6: $\mathcal{L}_{Stu}.\text{backward}()$
- 7: **Optimize** $\mathcal{O}_S$: $\mathcal{O}_S.\text{step}()$

Table 1

Comparison of unsupervised depth estimation Model Architectures(Inferencing on Titan XP with a 640×192 input size and batch size of 1).

Models	Params(MB)	Inference time(s)	a<1.25	a<1.25 ²	a<1.25 ³
Monodepth2	14.84	1.009×10^{-2}	0.877	0.959	0.981
HRDepth	14.61	1.466×10^{-2}	0.884	0.962	0.983
DIFFnet	10.87	7.831×10^{-2}	0.896	0.965	0.983
RA-Dpeth	9.98	8.290×10^{-2}	0.903	0.968	0.985
DPT-G	128.47	6.172×10^{-2}	0.905	0.970	0.986

and β is set to 0.001 in our experiments, $\tilde{M}_{s \rightarrow t}^{TOFM}$ is cropped from the $M_{s \rightarrow t}^{TOFM}$ in the same way like the $\tilde{D}_t^{Stu}$. Moreover, the smoothness term $\mathcal{L}_{sm}$ will be the same in Monodepth2 [3]. Due to the optimization of the flow decoder and the pose with two separate optimizers, it is necessary to use ‘detach’ between the semantic encoder and the features of the flow decoder to break the gradient connections.

4. Experiments

In this section, we first conducted tests on our TSUDepth using diverse network architectures and varying input image sizes. We selected Monodepth, HRDepth, and DIFFnet for evaluation, as our emphasis lies in innovating within the method framework, and transitioning to different network architectures resulted in superior outcomes. Simultaneously, to underscore the significant enhancement of small-resolution depth estimation accuracy by our CRED method, we performed tests at three distinct image resolutions: 416×128 , 640×192 , and 1024×320 . Secondly, to illustrate the performance improvements of our three proposed methods – TOFM, TSOO, and CRED – we conducted ablation experiments using the sophisticated DIFFnet as the test network. Additionally, we specifically carried out ablation experiments comparing our methods with RA-Depth. Thirdly, we showcased the performance of our proposed transformer network structure, DPT-G, using pre-trained models and including the GRU structure. Finally, we presented the performance of the TSUDepth framework on different datasets, accompanied by visualizations of depth images.

4.1. Implementation details

We conducted experiments on our method using two unsupervised monocular depth estimation datasets: KITTI [42] and Cityscapes [43]. KITTI comprises 39,810 training triplets, with 697 for evaluation. Cityscapes, designed for autonomous driving, provides 69,731 monocular triplets from the LeftImg8 bit Sequence for training. Additionally, we evaluated our model on Make3D [42].

Our training was implemented using PyTorch [44] with an NVIDIA RTX 3090Ti (24 GB). The input/output resolutions were set to 416×128 , 640×192 , and 1024×320 . We used ADAM [45] as the optimization algorithm, with other parameters set to $(\beta_1, \beta_2) =$

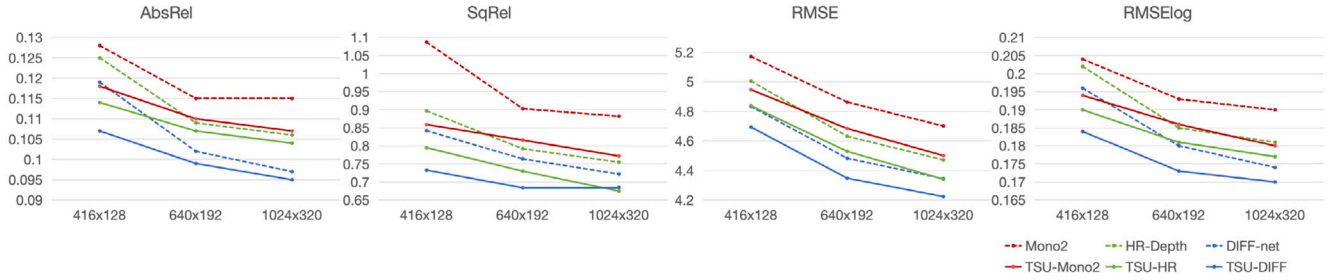


Fig. 12. Our method (marked by solid lines) can significantly improve the accuracy of the model on small-resolution inputs, models trained with different input sizes. The vertical axis denotes error metrics (the lower, the better).

Table 2

Ablation study on the adaptability of TSUDepth to various model structures. Results using the KITTI Eigen split. Our TSUDepth does not utilize post-processing (PP)[15]. Model* signifies construction through the transformer architecture.

Name	Model	Resolution	Error metric				Accuracy metric		
			AbsRel	SqRel	RMSE	RMSElog	a<1.25	a<1.25 ²	a<1.25 ³
Monodepth2[3]	Monodepth2	416 × 128	0.128	1.087	5.171	0.204	0.855	0.953	0.978
HR-Depth [4]	HR-Depth		0.125	0.897	5.005	0.202	0.855	0.954	0.980
DIFFNet [7]	DIFFNet		0.119	0.842	4.833	0.196	0.867	0.957	0.981
TSUDepth	Monodepth2		0.118	0.859	4.947	0.194	0.863	0.956	0.981
	HR-Depth	640 × 192	0.114	0.795	4.838	0.190	0.865	0.958	0.983
	DIFFNet		0.107	0.733	4.694	0.184	0.878	0.962	0.983
	DPT-G*		0.100	0.696	4.484	0.177	0.895	0.966	0.984
	Monodepth2		0.115	0.903	4.863	0.193	0.877	0.959	0.981
Monodepth2[3]	Monodepth2	640 × 192	0.109	0.792	4.632	0.185	0.884	0.962	0.983
HR-Depth [4]	HR-Depth		0.102	0.764	4.483	0.180	0.896	0.965	0.983
DIFFNet [7]	DIFFNet		0.101	0.729	4.454	0.178	0.897	0.965	0.983
Lite-Mono [36]	Lite-Mono*		0.099	0.708	4.372	0.175	0.900	0.967	0.984
MonoViT [41]	MonoViT*	1024 × 320	0.110	0.816	4.684	0.186	0.881	0.963	0.983
TSUDepth	Monodepth2		0.107	0.730	4.530	0.181	0.883	0.963	0.984
	HR-Depth		0.099	0.684	4.348	0.173	0.898	0.967	0.985
	DIFFNet		0.095	0.605	4.192	0.166	0.905	0.970	0.986
	DPT-G*	1024 × 320	0.115	0.882	4.701	0.190	0.879	0.961	0.982
Monodepth2[3]	Monodepth2		0.106	0.755	4.472	0.181	0.892	0.966	0.984
HR-Depth [4]	HR-Depth		0.097	0.722	4.345	0.174	0.907	0.967	0.984
DIFFNet [7]	DIFFNet		0.097	0.710	4.309	0.174	0.905	0.967	0.984
Lite-Mono [36]	Lite-Mono*	1024 × 320	0.096	0.714	4.292	0.172	0.908	0.968	0.984
MonoViT [41]	MonoViT*		0.107	0.772	4.501	0.180	0.886	0.964	0.984
TSUDepth	Monodepth2		0.104	0.675	4.341	0.177	0.890	0.965	0.985
	HR-Depth		0.095	0.685	4.224	0.170	0.906	0.968	0.985
	DIFFNet	1024 × 320	0.092	0.578	4.062	0.165	0.912	0.971	0.986
	DPT-G*		0.107	0.772	4.501	0.180	0.886	0.964	0.984
Monodepth2[3]	Monodepth2		0.104	0.675	4.341	0.177	0.890	0.965	0.985
HR-Depth [4]	HR-Depth		0.095	0.685	4.224	0.170	0.906	0.968	0.985
DIFFNet [7]	DIFFNet		0.092	0.578	4.062	0.165	0.912	0.971	0.986
Lite-Mono [36]	Lite-Mono*	1024 × 320	0.107	0.772	4.501	0.180	0.886	0.964	0.984
MonoViT [41]	MonoViT*		0.104	0.675	4.341	0.177	0.890	0.965	0.985
TSUDepth	Monodepth2		0.095	0.685	4.224	0.170	0.906	0.968	0.985
	HR-Depth		0.092	0.578	4.062	0.165	0.912	0.971	0.986
	DPT-G*		0.107	0.772	4.501	0.180	0.886	0.964	0.984

(0.9, 0.999). The SSIM weight was set to $\alpha = 0.85$, and the smoothness weight was set to $\lambda = 0.001$. A single-scale depth was used in the outputs following [6]. The learning rate for Resnet, HR-Depth, and DIFFNet was set to 1×10^{-4} , consistent with Monodepth2. The batch size was 8 with one GPU. The learning rate decayed to 1×10^{-5} after 15 epochs. Pre-trained checkpoints on ImageNet were employed. For the transformer-based DPT-G, the encoder's learning rate was set to 1×10^{-5} , and the decoder's to 1×10^{-4} [12]. The batch size was 16 with four GPUs.

4.2. Ablation study

From Table 1 (logging in our hardware and software environment, and the inference time is calculated as the mean over 100 iterations), we can observe the evolution of models from Monodepth2, based on a pure Resnet, to HRDepth based on a combination of Resnet [46] and HR-net [47], and finally to DIFFnet based on a pure HR-net. While the model parameters decrease, the complexity of the model increases, leading to an sharply increase in inference time. We selected CNN models incorporating both Resnet, HR-net structures (Monodepth2, HRDepth, DIFFNet) and ours DPT-G to validate our TSUDepth at different input sizes (416×128 , 640×192 , and 1024×320), as shown in Table 2. From Table 2, we can observe that, first, when comparing the same model structure and input size, our TSUDepth exhibits significant

performance improvements. Second, when comparing the performance of the same model structure at different resolutions, our TSUDepth framework demonstrates a remarkable enhancement in performance, particularly for low-resolution inputs. Moreover, when compared to Transformer-based methods, our TSUDepth framework, coupled with CNN-based DIFFnet, can approach or even surpass the Transformer-based Lite-Mono [36] and MonoViT [41] methods at various scales. This shows the outstanding performance of our approach.

To test the effectiveness of the proposed methods, we conducted ablation experiments, individually excluding TSOO, TOFM, and CRED. Additionally, we evaluated the substitution of DPT-G with DIFFnet in Table 4.

The Temporal Optical Flow Masking. When replacing Auto-Masking with TOFM, we observed improvements in various metrics. This is attributed to TOFM capturing the essence of unsupervised depth estimation by discerning distances through pixel motion sensitivity. As a result, minor movements no longer adversely affect depth estimation. Moreover, TOFM exhibits greater regional continuity than the Auto-Masking method, as illustrated in Fig. 3. This enhancement significantly preserves continuous supervisory information.

Temporal Symmetry-based Occlusion Optimization. We implemented the TSOO method replacing the MiniRL to achieve a precise

Table 3SOTA results on the Eigen Split with single 640×192 input.

Name	Error metric				Accuracy metric		
	AbsRel	SqRel	RMSE	RMSElog	a<1.25	a<1.25 ²	a<1.25 ³
SfMLearner [16]	0.183	1.595	6.709	0.270	0.734	0.902	0.959
SGDDepth [48]	0.113	0.835	4.693	0.191	0.879	0.961	0.981
SAFENet [49]	0.112	0.788	4.582	0.187	0.878	0.963	0.983
PackNet-SfM [17]	0.111	0.785	4.601	0.189	0.878	0.960	0.982
CADDepth [5]	0.105	0.769	4.535	0.181	0.892	0.964	0.983
Monodepth2[3]	0.115	0.903	4.863	0.193	0.877	0.959	0.981
HR-Depth [4]	0.109	0.792	4.632	0.185	0.884	0.962	0.983
RM-Depth [10]	0.108	0.710	4.513	0.183	0.884	0.964	0.983
tri-depth [19]	0.105	0.745	4.530	0.181	0.888	0.965	0.984
FSRE-Depth [18]	0.102	0.675	4.393	0.178	0.893	0.966	0.984
DIFFNet [7]	0.102	0.764	4.483	0.180	0.896	0.965	0.983
Lite-Mono(8M)[36]	0.101	0.729	4.454	0.178	0.897	0.965	0.983
MonoViT [41]	0.099	0.708	4.372	0.175	0.900	0.967	0.984
RA-Depth [6]	0.096	0.632	4.216	0.171	0.903	0.968	0.985
TSUDepth	0.095	0.605	4.192	0.166	0.905	0.970	0.986

Table 4Ablation study on the Eigen Split with single 640×192 input. w/o: without. w/o DPT-G: Switched to DIFFNet from DPT-G.

Ablation	Error metric				Accuracy metric		
	AbsRel	SqRel	RMSE	RMSElog	a<1.25	a<1.25 ²	a<1.25 ³
TSUDepth	0.095	0.605	4.192	0.166	0.905	0.970	0.986
TSUDepth w/o TSOO	0.101	0.724	4.587	0.177	0.900	0.967	0.984
TSUDepth w/o TOFM	0.096	0.655	4.458	0.170	0.903	0.969	0.985
TSUDepth w/o CRED	0.096	0.642	4.380	0.168	0.904	0.969	0.985
TSUDepth w/o DPT-G	0.099	0.684	4.348	0.173	0.898	0.967	0.985

identification of occluded regions. Compared to the Minimum Reprojection method [3], TSOO demonstrates fewer misjudgments. Additionally, it incorporates a linear combination of reprojection errors from multiple frames, resulting in a noticeable improvement.

The Cross-resolution Distillation Method. Our comparisons revealed that employing the CRED method can significantly enhance performance metrics. This is attributed to the substantial impact of input image size on depth estimation effectiveness, as evident from Table 2. With the same model and method, larger input image sizes result in higher accuracy in depth estimation, which can also be seen in Fig. 12. Exploiting this insight, we employed a distillation approach that leverages high-resolution information to assist in estimating depth from low-resolution inputs, yielding a notable improvement in effectiveness.

4.3. Depth estimation performance

We have achieved state-of-the-art performance on the Eigen Split dataset, as illustrated in Table 3. The visualizations of depth estimation, as depicted in Fig. 10, demonstrate that our method excels in producing clear and accurate depth estimations. Our model underwent testing on the Make3D dataset, as outlined in Table 6, having been trained on the KITTI dataset. Furthermore, we evaluated our model's performance on the Cityscapes dataset, as presented in Table 5. Cityscapes dataset presents challenges for unsupervised depth estimation, characterized by urban road scenes with frequent non-rigid movements of pedestrians and vehicles. Visualized in Fig. 13, our results show a closer alignment with the ground truth, clearer object boundaries, and superior metrics, as elaborated in Table 6. As shown in Table 1, our TSUDepth model achieves state-of-the-art (SOTA) results. However, it is undeniable that in terms of parameter count, the Transformer structure is not advantageous. Nevertheless, its inference speed (with a batch size of 1) is still faster than both DIFFNet and RA-Depth based on HR-Net. This is attributed to the complex multi-scale feature exchange in HR-Net [47] causing slower inference speeds.

Table 5Experiments on Cityscapes [43], with size 640×192 .

Name	Error metric				Accuracy metric		
	AbsRel	SqRel	RMSE	RMSElog	a<1.25	a<1.25 ²	a<1.25 ³
Struct2Depth [50]	0.145	1.737	7.280	0.205	0.813	0.942	0.976
GLNet [51]	0.129	1.044	5.361	0.212	0.843	0.938	0.976
Gordon et al.[52]	0.127	1.330	6.960	0.195	0.830	0.947	0.981
Li et al.[53]	0.119	1.290	6.980	0.190	0.846	0.952	0.982
Lee et al.[54]	0.111	1.158	6.437	0.182	0.868	0.961	0.983
RM-Depth [10]	0.090	0.825	5.503	0.143	0.913	0.980	0.993
TSUDepth	0.090	0.797	5.397	0.139	0.914	0.981	0.994

Table 6Experiments on Make3D [55], with size 640×192 .

Name	Error metric				Accuracy metric		
	AbsRel	SqRel	RMSE	RMSElog	a<1.25	a<1.25 ²	a<1.25 ³
Monodepth2	0.322	3.589	7.418	0.163	0.559	0.799	0.909
HR-Depth	0.315	3.208	7.024	0.159	0.562	0.798	0.913
DIFFNet	0.309	3.313	7.008	0.155	0.575	0.815	0.919
RA-Depth	0.277	2.703	6.548	0.143	0.612	0.846	0.933
TSUDepth	0.256	2.418	6.288	0.137	0.645	0.861	0.936

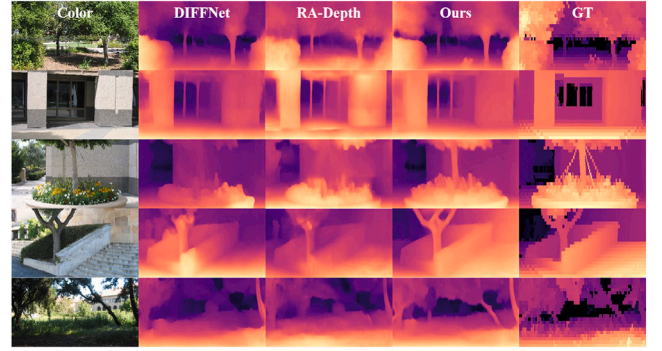


Fig. 13. Visualization of the depth estimation results on the Make3D dataset with a model trained on the KITTI dataset. It can be observed that ours produces closer depth results to GT than others, especially in the area of detailed structures (e.g., buildings, parterre, and tree trunks).

5. Conclusion

In this paper, we introduce a novel framework TSUDepth, designed to tackle challenges in unsupervised depth estimation based on photometric reprojection. TSUDepth incorporates the Temporal Symmetry-based Occlusion Optimization method (TSOO), leveraging uncertainty and fusing different frame errors via temporal symmetry to address object occlusion and visual shrinking occlusion issues. The Temporal Optical Flow Masking (TOFM) method predicts temporal optical flow and identifies stationary scene pixels, thereby mitigating misjudgments resulting from lack of motion or non-rigid motion. The Cross-resolution Distillation Method (CRED) enhances depth map prediction accuracy for smaller scales by utilizing a Teacher Depth Net to generate depth patches at the original resolution for supervising the output of the Student Depth Net at lower resolutions. Additionally, we propose a transformer-based model based on DPT, enhanced by a lightweight ConvGRU-based decoder, resulting in an efficient DPT-G model. Our experimental results validate the effectiveness of the proposed framework, demonstrating improved accuracy in estimating fine details of a scene under challenging illumination conditions.

CRediT authorship contribution statement

Yufan Zhu: Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation. **Rui Ren:** Writing – original draft, Visualization, Validation, Software. **Weisheng Dong:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Xin Li:** Writing – review & editing, Writing – original draft, Supervision, Methodology. **Guangming Shi:** Writing – review & editing, Supervision, Resources, Investigation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Weisheng Dong reports financial support was provided by the Natural Science Foundation of China. Xin Li reports financial support was provided by National Science Foundation. Co-author Xin Li previously employed by West Virginia University. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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